



LUKMAN HAKIM (he/him)

Nature-based Solutions Geospatial Professional

M.Sc. Photogrammetry and Geoinformatics | DAAD-EPOS Scholarship Awardee

Indonesian | Stuttgart, Germany | +49 176 86658462 | hakim.lukman1k@gmail.com | [Website](#)

PROFILE

Geospatial professional with 6+ years of experience supporting environmental, conservation, and public-sector geospatial projects. Experienced in developing GIS and remote-sensing workflows for spatial prioritization conservation zoning, ecosystem monitoring, mangrove mapping, benthic habitat mapping, land-cover analysis, and carbon-related assessments, including REDD+ and blue carbon initiative. Skilled in QGIS, ArcGIS Pro, Google Earth Engine, Python/Pandas, PostGIS, spatial data quality assurance, and reproducible geodata workflows. Strong background in integrating satellite imagery, field survey data, participatory mapping, administrative datasets, and open-source geospatial products into analysis-ready datasets and decision-support maps. Highly motivated to implement geospatial analysis for nature-based solutions, climate-focused planning, and environmental protection.

CORE COMPETENCIES

Remote Sensing & Spatial Modelling

Aerial Imagery · Satellite Imagery · LiDAR · Google Earth Engine · ENVI · SNAP · Image classification & quantification · Machine learning

Data Collection and Management

GNSS · Total Stations · Field Survey · UAS · Point Cloud · SQL · PostgreSQL/PostGIS · Relational Model · Metadata documentation

Programming & Automation

Python · TensorFlow · Rasterio · Pandas · PyTorch · API-based data processing · Automation · Reproducible workflows

GIS & Spatial Analysis

QGIS · ArcGIS Pro · Trimble · Cartography · CRS · Route Mapping · Conservation planning · Zoning support · Environmental monitoring

WebGIS & Geospatial Services

GeoServer · GeoDjango · Leaflet · Flask · JavaScript · JSON · REST APIs · OGC services

Tools & Collaboration

Git · Docker · FME · Technical documentation · Interdisciplinary teamwork · Stakeholder communication · Capacity Building

PROFESSIONAL EXPERIENCE

Hochschule für Technik Stuttgart — Akademischer Mitarbeiter / Research Associate

(08/2026 – Present)

Research project in collaboration with the Landesamt für Geoinformation und Landentwicklung Baden-Württemberg (LGL-BW)

- Extending previous research on automated green-roof detection in Baden-Württemberg by integrating high resolution aerial imagery with LoD2 building and roof-plane data for fine-scale vegetation mapping.
- Investigating deep-learning-based semantic segmentation approaches, including U-Net architectures, for delineating vegetated roof areas at higher spatial detail.
- Evaluating segmentation results against previous machine-learning approaches based on PlanetScope imagery and Random Forest classification, with emphasis on accuracy, reproducibility, and spatial quality control.
- Supporting the development of a scalable, high-resolution green-roof inventory for applications in urban climate adaptation and green-infrastructure assessment for official Baden-Württemberg State inventory.

Landesamt für Geoinformation und Landentwicklung Baden-Württemberg (LGL-BW) — Student Job

(06/2025 – 11/2025)

Identification of green roof areas in Baden-Württemberg using LoD2 building data and PlanetScope imagery

- Processed LoD2 building geometry (slope, aspect, area, height), semantic building attributes, and PlanetScope satellite imagery to support a 2025 roof-plane-level green roof inventory for Baden-Württemberg.
- Developed Python workflows for geodata preparation, feature extraction, quality control, and machine-learning-ready spatial datasets.
- Applied Random Forest classification to identify vegetated roof areas using spectral, geometric, and building-related features.

The Nature Conservancy Indonesia — GIS & Remote Sensing Specialist

(07/2019 – 08/2024)

- Supported cross-NGO and government collaborations in designing Marine Protected Area networks using MARXAN, gap analysis, and spatial prioritization to inform local, national, and transboundary zoning aligned with Aichi biodiversity conservation targets.
- Developed GIS and remote-sensing workflows for mangrove mapping, including distribution, canopy density, and aboveground biomass estimation to support forest carbon inventory, social forestry initiatives, and REDD+ readiness in Indonesia.
- Developed remote-sensing workflows for benthic habitat classification, including coral, seagrass, algae, sand, rubble, and rock classes, supporting blue carbon initiative
- Applied multi-temporal satellite datasets, including Maxar, SPOT, Landsat, Sentinel, PlanetScope, and Planet NICFI, using Google Earth Engine for environmental monitoring, land-cover analysis, and spatial modelling.
- Planned and conducted aerial data acquisition using small UAS drones and fixed-wing platforms, including flight planning, image triangulation, orthophoto correction, image stitching, and land-use/land-cover classification for coastal community landscapes.

- Integrated heterogeneous geospatial datasets from satellite imagery, field surveys, participatory mapping, administrative boundaries, and open-source geospatial products into structured spatial databases for analysis and decision-making.
- Mentored junior staff in GIS, remote sensing, data management, spatial analysis, and reproducible mapping workflows, improving internal technical capacity across conservation projects.

Research Institute for Tuna Fisheries Bali — Internship

(08/2018 – 09/2018)

- Collected fisheries-related spatial and environmental data for bachelor thesis research.
- Applied remote sensing and Generalized Additive Model (GAM) methods for distribution mapping of Yellowfin Tuna (*Thunnus albacares*) in the Indian Ocean.

EDUCATION

[Master of Science \(M.Sc.\), Photogrammetry and Geoinformatics — Hochschule für Technik Stuttgart](#)

(2024 – 2026, DAAD-EPOS Scholarship Awardee)

[Bachelor of Science \(B.Sc.\), Cartography and Remote Sensing — Universitas Gadjah Mada](#)

(2014 – 2019, BIDIKMISI Scholarship Awardee)

LANGUAGES

Javanese: Native | Indonesian: Native | English: C1 | German: B1-B2 | Japanese: Novice

SELECTED PUBLICATIONS

- Aung Kyaw, H., Ayala Rico, J., Awontemi, I. A., **et al.** (2026). [Urban Vegetation Mapping from High-Resolution Aerial Imagery with Deep Learning](#). DGPF Jahrestagung 2026.
- Fajariyanto, Y., Green, A., **Hakim, L.**, et al. (2024). [A Resilient Marine Protected Area Network Design: A First for the Arafura–Timor Seas](#). *Coastal Management*, 52(3), 147–173.
- Prakoso, D. A., **Hakim, L.**, et al. (2023). [The Dynamics of Mangrove and Pond Changes in East Kalimantan, Indonesia](#). *IOP Conference Series: Earth and Environmental Science*.
- Fajariyanto, Y., Green, A. L., Ramadyan, F., Suardana, N., **Hakim, L.**, et al. (2019). [Designing a Network of Marine Protected Areas for Fisheries Management Area 715 and Six Associated Provinces in Indonesia](#). Technical report prepared for the USAID Sustainable Ecosystems Advanced Project.
- **Hakim, L.**, Lazuardi, W., et al. (2018). [Assessing WorldView-2 Satellite Imagery Accuracy for Bathymetry Mapping in Pahawang Island, Indonesia](#). *IOP Conference Series: Earth and Environmental Science*.

ADDITIONAL QUALIFICATIONS

Small UAS Remote Pilot · Open Water Diver · Field-based coastal and marine survey experience

EXTRACURRICULAR

2023	Presenter at 5 th Asia Pacific Coral Reef Symposium, NUS Singapore
2023	Participant at Mangrove Monitoring International Training (MMIT), Archipelagic Island State Forum Bali Indonesia
2017	Presenter at World Blue Carbon Conference, Jakarta Convention Center Indonesia
2017	Presenter at Seminar Nasional Geomatika, Geospatial Information Agency Indonesia

Additional early experience includes serving as Vice President of the Geography Study Club at Universitas Gadjah Mada, and winning national college level debate competitions.

REFERENCES

- Prof. Dr. rer. nat. Michael Mommert | Hochschule für Technik Stuttgart | michael.mommert@hft-stuttgart.de
- Maryse Hilemann | Landesamt für Geoinformation und Landentwicklung Baden-Württemberg | Maryse.Hilemann@lgl.bwl.de
- Yusuf Fajariyanto | The Nature Conservancy Indonesia | yfajariyanto@tnc.org